

- **Identifying target use cases:** However good a memory and configuration a smartphone or tablet has, we must not forget that mobile applications are different from desktop applications in terms of screen size, memory, battery power and CPU cycles. The better these aspects are addressed by your app, the more it will be preferred by users. Hence, decide the use cases you want to extend to the mobile, keeping in view the following points:
 - (a) Consider if the use case really makes a strong reason to become part of the mobile app.
 - (b) The information shown or taken from the user should optimally utilise the view-port without cluttering it.
 - (c) The amount of information stored on the device should be purely from the perspective of offline mode usage. Any extra details should be downloaded from the server on a need basis and removed after use.
- **Offline usage support:** The application use cases should be evaluated from the perspective of all the key actors for the real scenarios. The scenarios highlighting off-network usage of the app should be analysed further and a plan for tackling those scenarios with identified restricted data access should be finalised.
- **Development approach:** Identify the target users of your application. If the application caters to the B2C segment, then the consumers would expect it to be supported on all popular mobile platforms. In such a case, app use cases should be analysed and, accordingly, the thin, hybrid or native application development approach should be selected.
- **Taking care of negative scenarios:** All negative scenarios should be analysed and taken care of in the app carefully.
- **Identify the target Android versions:** The Android minimum platform level should be targeted keeping in view the target Android versions for the app. The lower the platform level, the more devices the app would be able to cater to. Usually, a target Android version of ≥ 4.0 should provide you a decent coverage.
- **Identify the target Android devices:** The Android devices in the market today mostly fall under the category of MDPI, HDPI, XHDPI and XXHDPI. Also, the form factors available vary from 7.62 - 25.4 cm (3 - 10 inches). In such a diverse portfolio, it becomes important to decide the specific range of form factors, the DPI and reference devices you would consider for your app.
- For a B2C segment app, you would want to target about five reference devices. Among those five, two smartphones should have MDPI and HDPI resolutions, respectively, and there should be a smartphone and two tablets with XHDPI and XXHDPI resolution, and form factors of 12.7, 17.7 and 25.4 cm (5, 7 and 10 inch).
- **Design guidelines:** The applications developed for Android must adhere to certain design guidelines as specified on the Google site. The application UI must adhere to these guidelines in order to be accepted in Google Play, and it is also generally recommended that developers ensure the app

provides a uniform user experience on all Android devices.

- **White-labelling enabled app:** If the use cases targeted as part of the app deem to support white-labelling without the need for app upgrade on the respective app stores, then provision of dynamic configuration in the application and dynamic app behaviour (depending on the dynamic configuration elements pushed from the server) should be supported.
- **Extensibility:** App design and development should always be done from the point of view of the extensibility of the app. Sachin Naik adds, "Developing applications for higher performance with less memory and power consumption is still a challenge. The other challenge is related to Android fragmentation. With many Android versions and skins from OEMs, it is a challenge to write compatible applications."

Show me the money!

And what does the lucrative Android development market offer in terms of remuneration? Industry experts say the remuneration depends on the skill sets of the developers. An entry level engineer gets anywhere between Rs 500,000 to Rs 800,000 per annum. For an expert in the field, it can be anywhere between Rs 20,00,000 to Rs 25,00,000 per annum.

Handy tips to stay ahead of the competition

At a time when there are innumerable Android app developers, what is the mantra to have that extra edge over others? Develop apps that solve a problem and do some research to understand what your users want, feels Amrit. "Use Android guidelines, best practices and design practices as much as possible. Another thing to keep in mind is that the user experience and UI are not aspects that one should try to deal with right at the end of the development process. Making apps look good and following standard design guidelines should be something that is worked at early in the project. This saves a lot of time re-factoring code just to fit the UI later in the project cycle," he says.

The first extra edge is the thought and application itself. Creating an application based on a specific problem and target segment will help a lot, feels Naik. "As visibility in the crowded Android market is an issue, developers need to invest in getting the initial download numbers. There are many companies like Flurry, which help in getting the initial numbers. Once you have the critical mass, the application starts picking up traction. Developers also need to spend time in creating excellent product videos and constantly talk to users through social media. Once initial traction is in place, they must approach influential bloggers for reviews. Developers with a good understanding of design and user interaction patterns are in demand. Write apps and publish them in the Play store. That will give a clear perspective on what to improve on," he says. 

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